Supplementary material for "From Words to Desired Images: Language-Specified Goal Views for Visual Servoing in Gaussian Splatting Scenes"

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Abstract

This document complements the paper with the photometric servo results in detail (failure statistics, the criteria that go with success, Mip-NeRF 360 and current views from another reconstruction), the 3-D quality of the lifted objects, the sensitivity to the parameters and the ground-robot variant in synthetic houses. Section, table and figure numbers prefixed with S refer to this document; the others refer to the paper.

S1 Photometric servo

The protocol is that of the paper (Sec. 4.1): offsets of $0.1\,d^\star$ and 5° (small) or $0.2\,d^\star$ and 10° (large) in three directions, 600 iterations at 320 px, success within $0.02\,d^\star$ and 1°.

Why the servo fails. Photometric servoing has a narrow convergence basin [2], and most failures come from it rather than from unreachable desired images (Table S2): only 2 of our 67 desired images are never reached in their 6 runs, success decreases as the initial offset grows, and failed runs stop in a local minimum of the photometric error (median MSE 1647 against 24 for successful runs; 62% of them end farther in position from the goal than they started). The width of the basin still depends on the desired image: with the same servo and the same offsets, our desired images are reached more often than those of B2 and B3 at both offsets and than those of B1 at small offsets. For B1, the desired image is a photograph while the current views are renders, which leaves a residual error even at convergence (median MSE 120). Replacing the

Table S1 Photometric servo success (%, LERF-OVS, same model, 1572 runs toward the 262 desired images of all methods) when the desired image passes or fails each criterion, and odds ratio of success from a logistic regression on the four criteria with one intercept per method, without and with the log goal distance. The standard errors are scaled by the dispersion of the per-image fit (2.5), since the 6 runs of an image are correlated (*: $p < 0.05$).

criterion of the desired image	success (%) passed	failed	odds ratio [95% CI] criteria only	with $\log d^\star$
centred	77	63	1.75 [1.12, 2.74]*	1.55 [0.97, 2.47]
size 0.15–0.9	68	67	0.98 [0.67, 1.42]	0.98 [0.67, 1.44]
whole in frame	70	65	0.79 [0.49, 1.30]	0.81 [0.48, 1.34]
visible $\geq$50%	71	65	1.18 [0.75, 1.86]	1.30 [0.81, 2.08]
$\log d^\star$ (per unit)				0.66 [0.54, 0.81]*

Table S2 How the photometric servo fails (LERF-OVS, current views from the same model unless stated): desired images never reached in their 6 runs, success for small and large initial offsets, and median final photometric error (grey-level MSE) of failed and successful runs. CapFrame's 6 failed queries have no desired image and are not counted here.

desired images	images never reached	success (%) small	large	final MSE (median) failed	succeeded
Ours	2/67	85	76	1647	24
Ours, other model	9/67	75	67	1326	145
B1 detected training photo	1/67	70	71	2404	120
B1, rendered at the photo's pose	1/67	70	70	2531	73
B2 CLIP render-and-score	9/67	59	56	2121	35
B3 CapFrame [1]	6/61	66	60	2184	46

Table S3 Photometric servo toward our desired images: success (%) for small / large perturbations and median final errors.

dataset	current views	runs	small	large	median final $t/d^\star$, rot. (°)
LERF-OVS	same model	402	85	76	0.002, 0.09
LERF-OVS	other model (LangSplat-V2 3DGS)	402	75	67	0.004, 0.22
Mip-NeRF 360	same model, 320 px render	156	46	51	0.058, 1.71
Mip-NeRF 360	same model, 4× supersampled	156	62	56	0.002, 0.15

photograph by the 3DGS render at the photo's pose lowers this residual (MSE 73) but not the success rate (70% for both offsets): B1's lower success comes from the viewpoint, not from the photo–render gap. A servo with a wider basin, multi-scale or initialised by features, would raise the success of every method.

Servoing across reconstructions. Fig. S1 shows desired images on Mip-NeRF 360 and Table S3 the servo results. When the current views come from another 3DGS model of the scene (tested for our desired images only), success drops by about 10 points (85 to 75% and 76 to 67%; per image $p = 0.018$ and $p = 0.026$) but stays above 65%: the desired image transfers to a camera that does not render with our model. Part of the drop is a small systematic offset rather than a convergence failure: 20 of the 118 failed runs end within $0.05\,d^\star$ and 2° of the goal. On Mip-NeRF 360,

Table S4 Quality of the lifted object per stage-1 mask source on the 23 LERF-OVS queries both groundings find (medians): precision is the fraction of lifted points within 3% of the instance extent from the ground-truth instance, recall the fraction of the instance covered by the lifted points. The ground-truth polygon is lifted from one frame, while the ground truth merges several, hence its recall below 1.

stage-1 masks	precision	recall	F1
Grounding DINO + SAM	0.89	0.65	0.66
LangSplat-V2	0.68	0.49	0.59
GT label polygon	1.00	0.84	0.90

rendering the current view at 320 px from models trained on photos up to 5000 px wide aliases strongly [3], while the desired image is downsampled from a full-resolution render: at the goal pose itself, the median grey-level MSE between the two is 2426 on the bicycle scene and 102–1333 on the others, against 13–44 on LERF-OVS. Supersampling the current view (rendered 4× larger and averaged, as a camera integrates over its pixels) lowers it to 211 (50–218 on the others). Success rises from 46/51% to 62/56% (18 runs gained, 2 lost; 12 desired images improve and 2 get worse, Wilcoxon $p = 0.005$) and the median final error falls from $0.058\,d^\star$ to $0.002\,d^\star$; the garden scene goes from 33% to 92% and the bicycle scene (grass and gravel) stays at 17%, each over 12 runs (2 queries).

S2 Quality of the lifted object

Measured in 3-D against the ground-truth instance on the 23 queries both groundings find (Table S4), the object lifted from SAM masks lies more on the object than the one lifted from LangSplat-V2 masks (median precision 0.89 against 0.68, higher for 16 of the 23 queries), with a similar coverage (median recall 0.65 against 0.49, higher for 12), so its centroid, normal and extent are less contaminated by the background; neither difference is significant on 23 queries (Wilcoxon $p = 0.17$ and 0.26).

S3 Sensitivity

Table S5 reports how often the chosen direction stays within 5° of the frozen one. Most changes keep it for most queries. The fill fraction and the safety margin set the distance (d_{req} and μF), and through it the visibility, the fit and the free-space test of every candidate, so they also change the direction: halving the margin keeps it for 50% (Mip-NeRF 360) and 75% (LERF-OVS) of the queries, and the changed directions move by a median 28°. Two changes are capped: the fill cannot exceed the largest fill (0.85, +42%) and the margin cannot exceed 1 (+11%), which would put the camera behind the free space. The thresholds of the normal rule (0.6, 0.25, 0.98), the largest fill and the elevation percentiles were not varied. Replacing the planarity switch by a continuous weight $w = 1/(1 + (\rho/0.2)^4)$ between the two priors keeps the same view in 100% (Mip-NeRF 360) and 78% (LERF-OVS) of the queries.

Table S5 Same viewpoint as with the frozen parameters (angle $< 5°$, % of queries) when one parameter changes (by ±50% unless capped), or with a continuous score instead of the planar / non-planar switch.

parameter change	Mip-NeRF 360	LERF-OVS
fill ϕ 0.3 (−50%)	69	58
fill ϕ 0.85 (+42%, max.)	73	64
margin μ 0.45 (−50%)	50	75
margin μ 1.0 (+11%, max.)	96	88
planarity threshold 0.1 (−50%)	92	79
planarity threshold 0.3 (+50%)	88	90
cov. σ_θ 7.5° (−50%)	81	66
cov. σ_θ 22.5° (+50%)	88	76
cov. $\sigma_{\log d}$ 0.2 (−50%)	69	67
cov. $\sigma_{\log d}$ 0.6 (+50%)	88	67
azimuth step 7.5° (−50%)	65	66
azimuth step 22.5° (+50%)	46	54
continuous prior weight	100	78
coverage floor 0.25	92	99
both	92	76

Table S6 Ground robot in 9 InteriorGS houses (Habitat-GS validation split, 93 text queries: the ObjectNav categories of each house). Percentage of queries whose desired image shows an ObjectNav object of the category (visible) with its centre near the image centre horizontally (centred, ≤ 15% of the width; a level camera cannot centre vertically), and photometric servo success with 3 controlled degrees of freedom (small / large floor offset). B1 shares our grounding, hence the same right-object rate.

method	right object	visible	centred (horiz.)	good view	servo success (3 DOF) small	large
all queries ($n = 93$)						
B1 survey view	34	38	18	18	62	21
B2 CLIP render-and-score	–	16	5	5	64	34
Ours (ground robot)	34	32	26	26	65	34
right object found ($n = 32$)						
B1 survey view	100	78	38	38	67	17
B2 CLIP render-and-score	–	28	6	6	66	42
Ours (ground robot)	100	72	69	69	68	46

S4 A ground robot in synthetic houses

The previous experiments use a free-flying camera. We also test a ground robot, whose camera stands at a fixed height (1.25 m), level, and moves on the floor: three degrees of freedom (v_x, v_z, ω_y) of the camera. The scenes are 9 synthetic InteriorGS houses [4] distributed as 3DGS models with Habitat-GS [5], in metres, with a navigation mesh and ObjectNav goal objects [6]. They have no training photographs: the robot first surveys the house with 200 level views on navigable floor points (farthest-point sampling, random heading), which play the role of the training views for grounding and coverage. Each ObjectNav category present in a house is a text query (93 queries, 27 categories); the ground truth is the set of object centres of the category (no boxes or masks), so only visibility and horizontal centring are scored, with looser tests than on

LERF-OVS: the object is *visible* if its centre projects into the image and the surface seen at that pixel (rendered depth) lies within 1 m of it, our grounding finds the *right object* if the lifted centroid lies within 1 m of an object of the category, and a desired image is a good view if it is visible and centred.

Only the candidate set changes. The 74 candidates of the free-flying camera become 24 horizontal directions around the object; along each, the camera stands on the navigable floor point (10 cm grid) closest to the required distance d_{req} (searched between 0.4 and 1.5 d_{req}), at the robot's height and yawed toward the object centre; the free-space test is replaced by reachability: the floor point must lie on the navigation mesh (its connectivity to the robot's start is not checked). The score S and the selection rule are unchanged. B1 is the survey view in which the object is detected; B2 renders 8 headings at every survey position and keeps the one with the highest CLIP similarity. The servo starts from the goal moved on the floor by $0.1\,d^\star$ or $0.2\,d^\star$ in a random horizontal direction and turned by 5° or 10° about the vertical, and controls only the three robot velocities (the columns of $\boldsymbol{L}_I$ for v_x, v_z and ω_y). The goal distance $d^\star$, which scales the offsets and the tolerance, is the floor distance to the object centroid for ours, the distance from the survey camera to the centroid for B1, and the rendered depth at the image centre for B2. A query without a desired image counts as failed runs.

Table S6 gives the results and Fig. S2 examples. Grounding is harder than on the real scenes: the detector finds an object of the category in 34% of the queries. Small rooms (bathrooms, closets) are seen by few survey views, and the detector then fires on look-alikes (e.g. a wall panel for a toilet). For one query ("floor lamp") the object is found but no floor point is reachable at the required distance, a failure of our selection. When the object is found, our views are good in 69% of the 32 queries (95% interval 51–82) against 38% (23–55) for the survey view in which it was detected (exact McNemar test, $p = 0.04$): the survey view shows the object as often (78% vs 72%) but rarely centres it. Over all 93 queries the difference, 26% against 18%, is not significant ($p = 0.23$), since grounding fails for most of them. CLIP render-and-score rarely shows an object of the category (16% of the queries). With three controlled degrees of freedom, the servo reaches about two thirds of the runs of every method from the small offset; from the large one, it reaches ours more often than the survey views (34% against 21%, Wilcoxon test on the per-query success, $p = 0.018$; 46% against 17% when the object is found, $p = 0.003$) and as often as the CLIP views (34%), which are mostly close views of a wall or furniture (median $d^\star$ 0.75 m, against 1.6 m for ours and 3.0 m for the survey views; the three use different definitions of $d^\star$, see above). Success is lower than with the free-flying camera on LERF-OVS, possibly because the large untextured walls and floors of the synthetic houses give little photometric gradient. The survey is a single random draw per house.

References

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Fig. S1 Desired images on Mip-NeRF 360 for the training photo (B1), CLIP render-and-score (B2) and our method, on selected queries where our view is better than or similar to the training photo (no query of the *bicycle* scene qualifies). The scenes have no object labels, so the views are not scored; the badge on our images gives the photometric servo success over its six runs (current view supersampled 4×). Results over all 26 queries are in Table S3; failure cases are discussed in the limitations of the paper.



Fig. S2 Desired images for a ground robot in InteriorGS houses (camera at 1.25 m, level): survey view in which the object was detected (B1), CLIP render-and-score (B2) and our selection restricted to the navigable floor. The first six rows are selected among the 16 queries where our view is a good view and the survey view is not; results over all 98 queries are in Table S6. Last row: a failure. The small lamp is hidden from every reachable candidate at the robot's height (visibility 0 for all), so that every score and the tie-break key are zero and the first candidate, which faces a wall, is kept; this happens in 4 of the 92 selections. The survey view, in which the lamp was detected, is a good view. ✓/✗: good view (object centre visible and horizontally centred).